

Learning target AD5 and AD4, version 1

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.

AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

_____(AD5)
Consider the function $f(x) = 2x^3 + 24x^2 - 54x + 7$.

1. Find the critical points using algebra, and draw a sign chart (aka labeled number line).
Use the sign chart to decide what kind of extremum happens at each critical number.

- _____(AD4)
2. Restricting the domain of this function to the interval $[-8, 2]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-10, 2]$?

Learning target AD5 and AD4, version 2

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.

AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

_____ (AD5)
Consider the function $g(x) = -2x^3 + 18x^2 + 42x - 100$.

1. Find the critical numbers using algebra, and draw a sign chart (aka labeled number line).
Use the sign chart to decide what kind of extremum happens at each critical number.

- _____ (AD4)
2. Restricting the domain of this function to the interval $[-4, 4]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-4, 10]$?

Learning target AD5 and AD4, version 3

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.

AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

_____(AD5)
Consider the function $r(x) = -2x^3 + 18x^2 + 42x + 33$.

1. Find the critical numbers using algebra, and draw a sign chart (aka labeled number line).

Use the sign chart to decide what kind of extremum happens at each critical number.

Remember, only critical numbers go on the sign chart!

_____(AD4)

2. Restricting the domain of this function to the interval $[-3, 4]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-3, 8]$?